

GeoConDiffusion: Inference-Time Geometry-Constrained Sampling for Video-Diffusion Novel View Synthesis

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Abstract—Video-diffusion models have recently been used for camera-indexed novel view synthesis from sparse input views, but temporal coherence learned from natural videos does not ensure multi-view geometric compatibility. During inference, unconstrained initial noise may shift the global scene layout, while cross-frame attention may propagate information from regions that are invisible or depth-inconsistent under the target camera, leading to ghosting, texture drift, and long-range inconsistency. We propose GeoConDiffusion, an inference-time geometry-constrained sampling strategy for frozen video-diffusion-based NVS backbones. Instead of treating coarse geometry as a deterministic rendering target, GeoConDiffusion uses it as uncertain validity evidence to regulate information inheritance during sampling. Specifically, frequency-mixed initialization anchors reliable low-frequency layout cues while preserving high-frequency stochastic details, and confidence-gated cross-view attention suppresses geometrically invalid feature propagation during denoising. Without retraining the denoiser or replacing the base synthesis pipeline, GeoConDiffusion improves view fidelity, reconstruction-based consistency diagnostics, and downstream 3D reconstruction stability under matched camera protocols.

Index Terms—Novel view synthesis, video diffusion, inference-time sampling, geometric consistency, confidence-aware attention

I. INTRODUCTION

NOVEL view synthesis from sparse input views is increasingly used as an interface between generative modeling and 3D scene understanding. In NeRF [1], 3D Gaussian Splatting (3DGS) [2], and recent feed-forward reconstruction pipelines [3]–[6], synthesized views are not only visual outputs, but may also serve as additional observations for estimating or refining a 3D representation. This use case imposes a stronger requirement than frame-wise realism. A generated sequence should remain compatible with a single underlying scene; otherwise, small view-dependent inconsistencies may be amplified during downstream optimization into floaters, duplicated structures, blurred surfaces, or unstable texture.

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Video diffusion models [7], [8] provide a strong prior for generating coherent image sequences. This has motivated recent attempts to adapt video diffusion to camera-indexed novel view synthesis, where the sequence dimension is interpreted as camera motion. However, temporal coherence and geometric coherence are not equivalent. Natural videos often preserve semantic identity and local appearance continuity over time, whereas camera motion changes visibility, introduces parallax, exposes previously unseen regions, and invalidates some cross-frame correspondences. A video denoiser must therefore decide not only what content should appear in a target view, but also which information from other views can be legitimately inherited under the target camera. When this decision is made only implicitly by the learned video prior, the generated sequence may remain smooth and visually plausible while violating multi-view compatibility. Typical failures include silhouette drift, ghosting near occlusion boundaries, texture swimming, and inconsistent background structures.

A natural way to address this issue is to introduce geometric evidence. Sparse reconstruction, depth estimation, and recent foundation geometry models can provide approximate surface projection, visibility, and depth-ordering cues from limited input views [4], [9]–[12]. Yet sparse-view geometry is rarely complete or locally reliable. Thin structures, reflective regions, occlusion boundaries, and poorly observed surfaces can lead to inaccurate depth, missing support, or false correspondences. For this reason, coarse geometry should not be treated as a deterministic rendering target for generative synthesis. Its more defensible role is to act as uncertain validity evidence: it can indicate where cross-view information is likely to be valid, while leaving unsupported or ambiguous regions to the generative prior.

This distinction exposes a sampling-level gap in video-diffusion-based NVS. Existing geometry-aware pipelines often use coarse geometry as conditioning, warping, rendering support, or trajectory guidance. These uses are helpful, but they do not directly control two critical decisions inside the sampling process. First, the target latent is typically initialized from unconstrained Gaussian noise, allowing the global scene layout to drift early in denoising before geometric evidence can stabilize the trajectory. Second, cross-frame attention may aggregate features from reference views that are not visible or depth-consistent for a local target region. These errors are not simply caused by insufficient model capacity. They reflect the absence of an explicit validity mechanism for deciding when information inheritance across views should be permitted.

We propose GeoConDiffusion, an inference-time geometry-constrained sampling strategy for frozen video-diffusion-based novel view synthesis. GeoConDiffusion does not aim to



Fig. 1: **Inference-time geometric validity control for video-diffusion NVS.** Camera-indexed video diffusion can produce locally plausible novel views, but unconstrained sampling may still propagate information across geometrically invalid regions under large viewpoint changes. GeoConDiffusion uses coarse geometry as uncertain validity evidence rather than as a deterministic rendering target. Reliable low-frequency layout cues are injected into the initial latent, and confidence-gated cross-view attention reduces invalid information flow during denoising. Unsupported regions remain controlled by the generative prior.

reconstruct a complete explicit 3D scene before synthesis, nor does it replace the base denoiser or require additional training. Instead, it converts a coarse scaffold into visibility and depth-consistency cues that regulate sampling. At the initial state, Frequency-Mixed Initialization (FMI) anchors reliable low-frequency layout evidence while preserving high-frequency stochastic components for appearance synthesis. During denoising, Confidence-Gated Cross-View Attention (CGA) suppresses feature propagation from geometrically unsupported reference views at local target regions. In this way, coarse geometry constrains where information may be inherited, without forcing the generated views to copy noisy scaffold renderings.

The paper makes three contributions. First, we formulate geometry-constrained video-diffusion NVS as a sampling-time validity problem: coarse geometry is used to judge information inheritance across views, rather than to provide a deterministic target image. Second, we introduce FMI, which biases the initial latent toward reliable low-frequency geometric evidence while retaining stochastic high-frequency details. Third, we introduce CGA, which uses visibility and depth agreement to reduce invalid cross-view feature propagation during denoising. Both components operate only at inference time and can be applied to a frozen video-diffusion NVS backbone.

We evaluate GeoConDiffusion under matched sparse-view inputs, target camera settings, and evaluation protocols. Across multiple datasets, the proposed sampling constraints improve image fidelity and perceptual quality compared with unconstrained sampling and representative NVS baselines. Reconstruction-based diagnostics further show lower pose inconsistency on generated sequences, and downstream 3DGS optimization produces more stable geometry under the same reconstruction protocol. These results indicate that coarse geometry can be more effectively used as sampling-time validity evidence than as a hard rendering target.

II. RELATED WORK

A. Diffusion Priors for Sparse-View Novel View Synthesis

Sparse-view novel view synthesis is highly under-constrained because unseen geometry, occluded appearance, and view-dependent effects must be inferred from limited observations. Diffusion models address this ambiguity by introducing strong generative priors learned from large-scale visual data. Early object-centric methods such as Zero-1-to-3 [13] condition a pretrained diffusion model on an input image and a relative camera transformation to synthesize novel object views. ZeroNVS [14] further reduces camera and scale ambiguity through field-of-view normalization. These methods demonstrate that pretrained diffusion priors can hallucinate plausible unseen content, but views generated independently or weakly coupled across cameras often lack strict multi-view compatibility.

Subsequent methods introduce stronger interaction among generated views or incorporate geometric cues into the generation process. SyncDreamer [15] synchronizes multi-view denoising through a 4D feature volume, while MultiDiff [16] uses monocular depth and warping to encourage cross-view consistency. Other works improve few-shot or camera-conditioned synthesis by refining view transitions or relaxing explicit geometric assumptions [17], [18]. Although these designs improve view consistency, the central challenge remains unresolved under large viewpoint changes: the generator must distinguish information that can be inherited from existing views from content that should be newly synthesized. This distinction is difficult to learn implicitly from appearance priors alone, especially around disocclusions, occlusion boundaries, and thin structures.

B. Video Diffusion for Camera-Indexed Generation

Video diffusion models provide a natural backbone for generating sequences with temporal coherence [7], [8]. Since a camera trajectory can be represented as an ordered sequence of

images, recent methods adapt video diffusion priors to camera-controlled synthesis and 3D-aware generation. SV3D [19] fine-tunes Stable Video Diffusion with camera conditioning for object-centric orbital views. MotionCtrl [20] and CameraCtrl [21] introduce camera-control mechanisms to guide video generation under specified motion. V3D [22] explores the potential of pretrained video models for 3D generation and view synthesis.

However, temporal coherence is not the same constraint as multi-view geometric coherence. Temporal video generation mainly requires plausible appearance evolution across adjacent frames, whereas camera-indexed generation must respect visibility changes, parallax, disocclusion, and depth ordering induced by viewpoint change. A sequence can therefore be temporally smooth but geometrically incompatible: textures may drift over surfaces, silhouettes may shift, and background structures may fail to correspond to a single 3D scene. This limitation is particularly important when generated views are used as observations for downstream reconstruction. Our work addresses this gap from the sampling side. Instead of relying on the video prior to implicitly learn all cross-view validity decisions, we use coarse geometric evidence to regulate when cross-view information can be propagated during denoising.

C. Geometry Scaffolds for View Synthesis and Reconstruction

Explicit geometry has long been used to constrain novel view synthesis. Generalizable reconstruction methods predict 3D representations from sparse observations, including NeRF-style radiance fields and recent Gaussian-splatting-based representations. pixelSplat [3] predicts 3D Gaussian primitives from image pairs, while MVSplat [4], TranSplat [5], DepthSplat [23], and Flash3D [6] incorporate multi-view, transformer, or depth-aware designs to improve sparse-view reconstruction. These methods provide explicit 3D outputs and can render geometrically grounded views, but their quality may degrade when the input views are extremely sparse, wide-baseline, or ambiguous.

Another line of work uses coarse geometry as an intermediate scaffold rather than as the final representation. Sparse reconstruction systems and foundation geometry models such as COLMAP [9], DUST3R [10], and VGGT [11] can estimate camera parameters, depth, or point-cloud structures from limited images. Such outputs provide useful visibility and depth-ordering cues for generative synthesis, but they are incomplete and locally unreliable in sparse-view settings. A scaffold may miss disoccluded surfaces, distort thin structures, or produce incorrect correspondences around reflective and textureless regions. Directly treating this scaffold as a rendering target can therefore over-constrain the generator or propagate geometric errors into the synthesized views.

GeoConDiffusion follows the scaffold-based view that coarse geometry is useful, but differs in how the scaffold is used. We do not require it to produce a complete or photorealistic 3D reconstruction. Instead, we convert scaffold-derived visibility and depth agreement into sampling-time validity cues. These cues affect the initial latent distribution and the cross-view attention flow of a frozen video-diffusion

NVS backbone. In this sense, the scaffold is used to decide where information inheritance is credible, while the diffusion prior remains responsible for synthesizing unsupported or uncertain regions.

III. METHOD

We present GeoConDiffusion, an inference-time sampling strategy for video-diffusion-based novel view synthesis. The goal is not to reconstruct a complete explicit 3D scene before synthesis, but to use coarse geometry to decide where cross-view information inheritance is reliable. As shown in Fig. 2, a coarse scaffold is first estimated from sparse input views and then converted into visibility and depth-consistency cues. These cues constrain two points in the sampling process of a frozen video-diffusion NVS backbone: the initial latent state and the cross-view attention flow. Reliable scaffold-supported regions are used to stabilize low-frequency layout and valid information propagation, while low-confidence or unsupported regions remain primarily controlled by the diffusion prior.

A. Problem Setup and Sampling Overview

Given a sparse set of input images

$$\mathcal{I} = \{I_r\}_{r=1}^R, \quad (1)$$

where R is the number of reference views, the task is to synthesize a sequence of target views

$$\hat{\mathcal{V}} = \{\hat{V}_n\}_{n=1}^N \quad (2)$$

under target camera poses

$$\hat{\Pi} = \{\hat{\pi}_n\}_{n=1}^N, \quad \hat{\pi}_n = (K_n, R_n, \mathbf{t}_n). \quad (3)$$

Here n indexes the target camera view, while k will be used for the diffusion timestep. This separation avoids mixing camera sequence indices with the denoising schedule. We use a latent video-diffusion backbone with a VAE encoder-decoder $E(\cdot), D(\cdot)$ and a frozen denoiser ϵ_θ . The reference image latent is written as

$$z_r = E(I_r). \quad (4)$$

In standard video-diffusion sampling, the target latent sequence is initialized from Gaussian noise and iteratively denoised by ϵ_θ . When the sequence dimension corresponds to camera motion, this unconstrained sampling procedure has two failure modes. First, the global layout may drift early because the initial latent contains no view-dependent geometric support. Second, cross-frame attention may aggregate features from reference frames that are not geometrically valid for a local target region. GeoConDiffusion addresses these failures by constructing geometry-derived validity evidence and using it to regulate initialization and attention during inference.

B. Geometry-Derived Validity Evidence

We first estimate a coarse geometric scaffold $\mathcal{P}$ from the sparse input views using an off-the-shelf geometry lifting module, such as sparse reconstruction or a foundation geometry model [9]–[11]. The scaffold is not assumed to be complete or photorealistic. It is only required to provide approximate

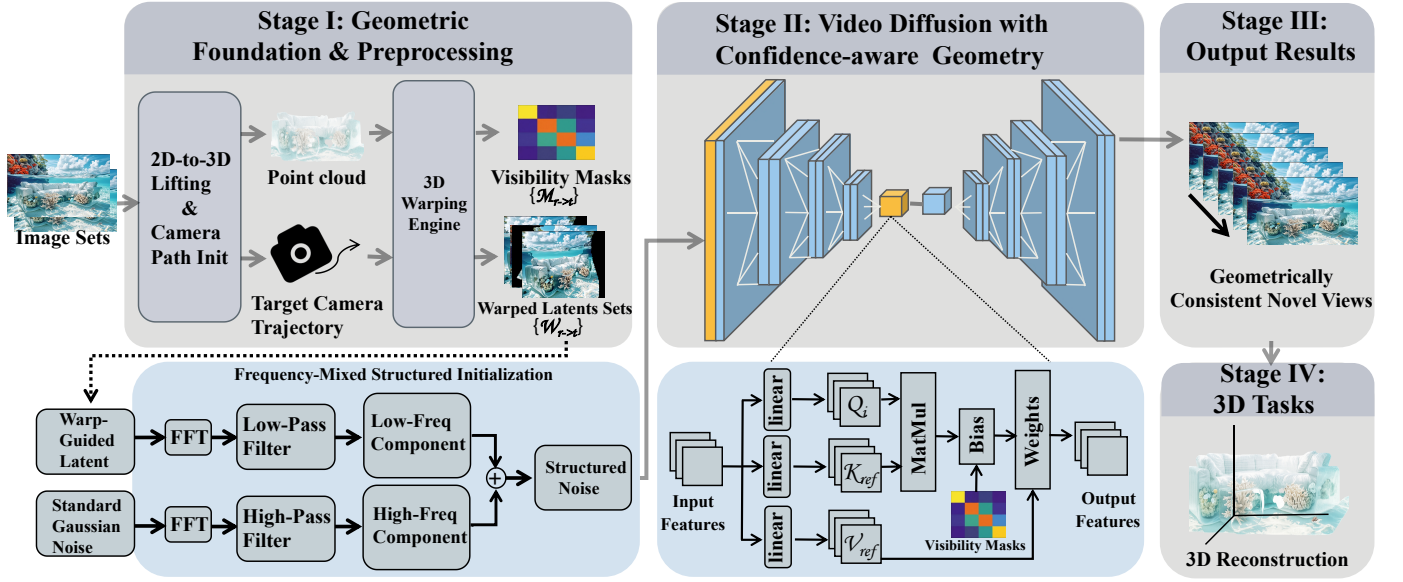


Fig. 2: **Overview of GeoConDiffusion.** GeoConDiffusion regulates the inference-time sampling process of a frozen video-diffusion NVS backbone with geometry-derived validity evidence. Given sparse input views, a coarse scaffold provides warped evidence, visibility masks, and depth-consistency confidence. These cues are injected at two sampling control points: Frequency-Mixed Initialization (FMI), which anchors reliable low-frequency layout evidence before denoising, and Confidence-Gated Cross-View Attention (CGA), which reduces invalid cross-view information flow during denoising. The scaffold is used as uncertain validity evidence rather than as a deterministic rendering target, so unsupported regions remain governed by the generative prior.

projection, visibility, and depth-ordering evidence for judging whether information from a reference view can be inherited by a target view.

For each target pose $\hat{\pi}_n$, we render the scaffold with a lightweight z-buffer renderer to obtain a coarse target depth map D_n . The corresponding image-space support mask is

$$M_n^{\text{img}}(p) = \mathbb{I}(D_n(p) > 0), \quad (5)$$

where p denotes an image pixel. Valid pixels are regions explained by the scaffold from the target viewpoint. Pixels outside this support are treated as disoccluded or unsupported by the current scaffold.

For a reference view r and a target view n , we reproject the reference observation into the target camera using the scaffold geometry. This produces warped evidence, a reprojection mask, and a reprojected depth map:

$$(\tilde{I}_{r \rightarrow n}, M_{r \rightarrow n}^{\text{img}}, \tilde{D}_{r \rightarrow n}) = \mathcal{W}(I_r, \mathcal{P}, \pi_r, \hat{\pi}_n), \quad (6)$$

where $\mathcal{W}(\cdot)$ denotes geometry-based warping. A pixel is considered valid only when its reprojection lies inside the target image boundary and passes visibility checks.

Because the scaffold may be incomplete or locally inaccurate, validity cannot be determined by reprojection support alone. We define a depth-consistency confidence map:

$$C_{r \rightarrow n}^{\text{img}}(p) = M_{r \rightarrow n}^{\text{img}}(p) \exp \left(- \frac{|\tilde{D}_{r \rightarrow n}(p) - D_n(p)|}{\tau} \right), \quad (7)$$

where τ controls the tolerance to depth disagreement. This confidence is high only when the reference-to-target repro-

jection is both visible and depth-consistent with the target scaffold.

When multiple reference views are available, their candidate supports are aggregated as

$$\bar{M}_n^{\text{img}} = \bigcup_{r=1}^R M_{r \rightarrow n}^{\text{img}}. \quad (8)$$

This union mask indicates where at least one reference view provides geometric support. It is not used for direct color averaging, since averaging warped evidence can blur details or propagate incorrect colors when the scaffold is noisy.

We then estimate the reliability of each reference-target pair. The overlap between reference r and target n is measured by

$$\text{IoU}(r, n) = \frac{|M_{r \rightarrow n}^{\text{img}} \cap M_n^{\text{img}}|}{|M_{r \rightarrow n}^{\text{img}} \cup M_n^{\text{img}}|}. \quad (9)$$

Since overlap alone cannot distinguish correct support from depth-inconsistent reprojection, we combine it with average confidence:

$$\rho(r, n) = \text{IoU}(r, n) \cdot \frac{\sum_p C_{r \rightarrow n}^{\text{img}}(p)}{\sum_p M_{r \rightarrow n}^{\text{img}}(p) + \varepsilon}, \quad (10)$$

where ε avoids numerical instability. For each target view n , we retain the top- m reference views whose reliability exceeds a threshold:

$$\mathcal{R}_n = \text{Top-}m(\{r \mid \rho(r, n) > \rho_{\min}\}). \quad (11)$$

The selected set $\mathcal{R}_n$ determines which reference views are allowed to support target view n , while $C_{r \rightarrow n}^{\text{img}}$ determines where such support is reliable.

C. Frequency-Mixed Initialization

Standard diffusion sampling initializes the target latent from Gaussian noise. For camera-indexed NVS, this can cause early layout drift because the initial latent has no geometric preference. A direct initialization from warped evidence is also undesirable, because the scaffold may be incomplete or wrong in disoccluded regions. We therefore introduce Frequency-Mixed Initialization (FMI), which anchors reliable low-frequency layout evidence while preserving high-frequency stochasticity for generative synthesis.

For each target view n , we select the most reliable reference view:

$$r_n^* = \arg \max_{r \in \mathcal{R}_n} \rho(r, n). \quad (12)$$

The warped evidence from this reference is encoded into the latent space:

$$\tilde{z}_{r_n^* \rightarrow n} = E(\tilde{I}_{r_n^* \rightarrow n}). \quad (13)$$

Let K denote the initial diffusion timestep. Following the forward diffusion process, we perturb the warped latent to the initial noise level:

$$z_{K,n}^{\text{geo}} = \sqrt{\bar{\alpha}_K} \tilde{z}_{r_n^* \rightarrow n} + \sqrt{1 - \bar{\alpha}_K} \eta'_n, \quad \eta'_n \sim \mathcal{N}(0, I), \quad (14)$$

where $\bar{\alpha}_K$ is the cumulative noise coefficient at timestep K .

We then mix the low-frequency spectrum of this geometry-aligned noisy latent with the high-frequency spectrum of fresh Gaussian noise:

$$\Phi_{K,n} = L \odot \mathcal{F}(z_{K,n}^{\text{geo}}) + (1 - L) \odot \mathcal{F}(\eta_n), \quad (15)$$

$$x_{K,n}^{\text{fmi}} = \mathcal{F}^{-1}(\Phi_{K,n}), \quad \eta_n \sim \mathcal{N}(0, I). \quad (16)$$

Here $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the Fourier transform and inverse Fourier transform applied along the spatial dimensions of the latent feature map, and L is a low-pass filter. This frequency split reflects the different roles of geometry and the diffusion prior: geometry is more reliable for coarse layout, silhouette, and perspective, while high-frequency texture and fine details should remain stochastic when scaffold evidence is uncertain.

To avoid over-anchoring unreliable regions, we construct a confidence-gated latent blending map. The strongest available confidence for target view n is

$$\hat{C}_n^{\text{img}}(p) = \max_{r \in \mathcal{R}_n} C_{r \rightarrow n}^{\text{img}}(p). \quad (17)$$

The image-space mask and confidence map are downsampled to the latent resolution:

$$\hat{M}_n^\ell = \text{Down}_\ell(\bar{M}_n^{\text{img}}), \quad \hat{C}_n^\ell = \text{Down}_\ell(\hat{C}_n^{\text{img}}), \quad (18)$$

where ℓ denotes the latent attention resolution.

The frame-level anchoring strength combines scaffold coverage and average confidence:

$$\alpha_n = \text{clip} \left(\frac{|\bar{M}_n^{\text{img}}|}{HW} \cdot \frac{\sum_p \hat{C}_n^{\text{img}}(p)}{\sum_p \bar{M}_n^{\text{img}}(p) + \varepsilon}, 0, \alpha_{\max} \right), \quad (19)$$

where H and W are the image height and width, and $\alpha_{\max}$ prevents excessive reliance on the scaffold. The latent anchoring map is then

$$A_n^\ell = \alpha_n \hat{C}_n^\ell \odot \hat{M}_n^\ell. \quad (20)$$

The final initial latent is obtained by confidence-gated blending:

$$x_{K,n} = (1 - A_n^\ell) \odot \eta_n + A_n^\ell \odot x_{K,n}^{\text{fmi}}. \quad (21)$$

This formulation does not copy scaffold renderings into the generation result. When the scaffold is sparse or unreliable, A_n^ℓ becomes small and the initialization approaches standard Gaussian noise. When scaffold support is reliable, the initial latent is biased toward a stable low-frequency layout while retaining stochastic high-frequency components for appearance synthesis.

D. Confidence-Gated Cross-View Attention

FMI stabilizes the initial latent state, but denoising still depends on cross-frame information propagation. If cross-frame attention freely attends to all reference frames, the denoiser may propagate features from views that are not geometrically valid for a local target region. This is especially harmful near occlusion boundaries, where invalid feature inheritance can produce ghosting, duplicated structures, or texture drift. We therefore introduce Confidence-Gated Cross-View Attention (CGA) to regulate cross-view attention during denoising.

For each selected reference-target pair (r, n) , we downsample the confidence map to the latent attention resolution:

$$C_{r \rightarrow n}^\ell = \text{Down}_\ell(C_{r \rightarrow n}^{\text{img}}). \quad (22)$$

Let token i be a query token located at p_i in target view n , and let token j be a key token from source view $s(j)$. We define the local inheritance validity of source view $s(j)$ for target token i as

$$g_{i,s(j)} = C_{s(j) \rightarrow n}^\ell(p_i) \cdot \mathbb{I}(s(j) \in \mathcal{R}_n). \quad (23)$$

This gate is conditioned on the target-token location. It estimates whether the source frame is geometrically valid for supporting the current target region, rather than enforcing a hard pixel-level correspondence between token i and token j .

Given query q_i , key k_j , and feature dimension d , the original attention logit is modified as

$$\text{sim}_{ij} = \frac{q_i^\top k_j}{\sqrt{d}} + B_{ij}, \quad (24)$$

where the geometry-derived attention bias is

$$B_{ij} = \begin{cases} \beta \log(g_{i,s(j)} + \delta), & s(j) \in \mathcal{R}_n, \\ -\Gamma, & s(j) \notin \mathcal{R}_n. \end{cases} \quad (25)$$

Here β controls the strength of the confidence bias, δ avoids numerical instability, and Γ suppresses unsupported cross-view interactions. The resulting attention weights are computed as

$$a_{ij} = \frac{\exp(\text{sim}_{ij})}{\sum_{j'} \exp(\text{sim}_{ij'})}. \quad (26)$$

CGA does not force the target token to copy a specific source token. Instead, it reduces attention mass from source views that are not visible or depth-consistent for the target region. Intra-frame attention and low-confidence regions remain available to the diffusion prior, allowing disoccluded or unsupported content to be synthesized rather than incorrectly inherited.

E. Inference Procedure

GeoConDiffusion modifies only the inference-time sampling process. The denoiser ϵ_θ , VAE encoder-decoder, and base camera-conditioned NVS pipeline remain frozen. For each target view, we first compute geometry-derived evidence using Eqs. (5)–(11). FMI then constructs the initial latent $x_{K,n}$ using Eqs. (12)–(21). During each denoising step, CGA modifies cross-view attention logits according to Eqs. (22)–(26). After the final denoising step, the synthesized view is decoded as

$$\hat{V}_n = D(x_{0,n}). \quad (27)$$

All target views are generated under the same specified camera protocol as the base NVS backbone. GeoConDiffusion therefore changes the sampling trajectory, not the training objective, network architecture, or target camera definition. This property makes it suitable for controlled comparison with the corresponding unconstrained sampler.

IV. EXPERIMENTS

We evaluate GeoConDiffusion from three perspectives. First, we measure whether the proposed inference-time constraints improve novel-view image quality under matched sparse-view inputs and target camera settings. Second, we use reconstruction-based diagnostics to examine whether the generated sequences are more compatible with a single camera-indexed scene interpretation. Third, we evaluate whether the synthesized views provide more stable observations for downstream 3DGS reconstruction under a fixed reconstruction protocol. Since GeoConDiffusion modifies only sampling and keeps the video-diffusion NVS backbone frozen, the main experimental question is not whether it constitutes a new end-to-end NVS architecture, but whether geometry-derived validity cues improve the behavior of an existing sampler.

A. Experimental Setup

a) Datasets.: We evaluate GeoConDiffusion on datasets covering indoor, outdoor, object-centric, and reconstruction-oriented settings. RealEstate10K [24] is used for indoor scenes with camera motion and large layout variation. CO3D [25] evaluates object-centric view synthesis under diverse categories and trajectories. Tanks-and-Temples [26] provides complex outdoor scenes with challenging geometry and background structures. DL3DV-10K [27] is used for additional qualitative evaluation on detailed indoor and outdoor layouts. For cross-domain and downstream analysis, we further report results on DTU [28], ACID [29], and Spring [30]. All evaluations use the same input views and target camera settings for methods that expose compatible camera-control interfaces.

b) Baselines.: We compare GeoConDiffusion with representative sparse-view, generative, and reconstruction-based NVS methods, including ZeroNVS [14], LucidDreamer [31], MotionCtrl [20], ViewCrafter [32], FreeSplat [33], Flash3D [6], and InstantSplat [34]. For DTU and ACID, we additionally include pixelNeRF [35], pixelSplat [3], MVSPat [4], TranSplat [5], and DepthSplat [23] following their corresponding evaluation settings. When a baseline does not support the same camera protocol or input configuration,

the comparison is treated as protocol-limited rather than strictly controlled. The most controlled comparison is between GeoConDiffusion and the same frozen video-diffusion NVS backbone without FMI and CGA.

c) Metrics.: We report standard image fidelity and perceptual metrics, including PSNR, SSIM, LPIPS [36], DISTS [37], and FID [38]. To diagnose cross-view compatibility, we run COLMAP [9] on generated sequences and compare the recovered camera poses with the target camera trajectory, reporting rotation error R_{dist} and translation error T_{dist} . These pose errors should be interpreted as reconstruction-based consistency diagnostics rather than direct ground-truth measurements of all geometric errors. For downstream 3D reconstruction, generated views are used as additional observations for 3DGS optimization, and the rendered views from the optimized 3D representation are evaluated using PSNR, SSIM, and LPIPS under the same reconstruction protocol.

B. Quantitative Results

Table I reports quantitative comparisons on Tanks-and-Temples, RealEstate10K, and CO3D. GeoConDiffusion improves the frozen video-diffusion NVS backbone and obtains competitive results against representative sparse-view, generative, and reconstruction-based baselines under the evaluated protocols. The gains are larger on hard splits, where large viewpoint changes and disocclusions make invalid cross-view information propagation more likely. In addition to image-level metrics, the reduction in R_{dist} and T_{dist} indicates that the generated sequences are more stable under the COLMAP-based reconstruction diagnostic.

The hard-split results are particularly informative because they contain larger viewpoint changes and more frequent disocclusions. On Tanks-and-Temples, GeoConDiffusion improves hard-set PSNR from 17.11 to 20.12 and reduces R_{dist} from 1.387 to 0.935 compared with ViewCrafter. On RealEstate10K, hard-set PSNR increases from 20.20 to 24.56, while R_{dist} decreases from 1.874 to 0.897. On CO3D, PSNR improves from 16.33 to 20.13, and T_{dist} decreases from 2.459 to 1.198. These trends are consistent with the intended role of FMI and CGA: reliable low-frequency geometry stabilizes early sampling, while confidence-gated attention reduces invalid cross-view feature propagation.

Table II evaluates whether the generated views are useful as observations for downstream 3D reconstruction. All methods are evaluated under the same downstream reconstruction protocol whenever applicable. On Tanks-and-Temples, GeoConDiffusion reaches 23.45 PSNR, 0.780 SSIM, and 0.190 LPIPS. On ACID, it obtains 21.45 PSNR, 0.784 SSIM, and 0.193 LPIPS. These results suggest that the proposed sampling constraints improve not only isolated view quality, but also the utility of generated views for subsequent 3DGS optimization under the fixed protocol.

Figure 4 visualizes representative generated sequences and their corresponding downstream 3DGS reconstructions. Reconstructions from GeoConDiffusion-generated views contain fewer visible artifacts and more stable scene structures, which is consistent with the quantitative results in Table II. This



Fig. 3: **Visualization of scaffold evidence and reconstruction feedback.** Sparse-view geometry models provide useful but incomplete scaffold evidence. GeoConDiffusion does not replace these lifting models or require a complete explicit reconstruction before synthesis. Instead, it uses scaffold-derived visibility and depth agreement to regulate sampling, and the resulting generated views can provide more stable observations for subsequent geometric processing.

Method	Easy set						Hard set					
	LPIPS ↓	PSNR ↑	SSIM ↑	FID ↓	R_{dist} ↓	T_{dist} ↓	LPIPS ↓	PSNR ↑	SSIM ↑	FID ↓	R_{dist} ↓	T_{dist} ↓
<i>Tanks-and-Temples dataset [26]</i>												
LucidDreamer [31]	0.413	14.53	0.362	42.32	6.137	5.695	0.558	11.69	0.267	200.8	8.998	9.305
ZeroNVS [14]	0.482	14.71	0.380	74.60	8.810	6.348	0.569	12.05	0.309	131.0	8.860	8.557
MotionCtrl [20]	0.400	15.34	0.427	70.30	7.299	8.039	0.473	13.29	0.384	196.8	9.801	9.112
ViewCrafter [32]	0.198	20.23	0.573	30.08	0.573	1.134	0.301	17.11	0.465	39.12	1.387	1.213
Ours	0.191	22.31	0.786	26.34	0.397	0.918	0.213	20.12	0.625	36.17	0.935	0.803
<i>RealEstate10K dataset [24]</i>												
LucidDreamer [31]	0.315	16.35	0.579	56.77	5.821	10.02	0.400	14.13	0.511	71.43	7.990	10.85
ZeroNVS [14]	0.364	16.50	0.577	96.18	6.370	9.817	0.431	14.24	0.535	105.8	8.562	10.31
MotionCtrl [20]	0.341	16.31	0.604	89.90	4.236	9.091	0.386	16.29	0.587	70.02	8.084	9.295
ViewCrafter [32]	0.163	19.65	0.694	37.26	0.483	3.653	0.265	20.20	0.675	26.76	1.874	3.345
Ours	0.132	22.76	0.883	32.53	0.354	2.125	0.156	24.56	0.924	23.36	0.897	2.190
<i>CO3D dataset [25]</i>												
LucidDreamer [31]	0.429	15.11	0.451	78.87	12.90	6.665	0.517	12.69	0.374	157.8	16.43	8.301
ZeroNVS [14]	0.467	15.15	0.463	93.84	15.44	8.872	0.524	13.31	0.426	143.2	15.02	10.22
MotionCtrl [20]	0.393	16.87	0.529	69.18	16.87	5.131	0.443	15.46	0.502	112.7	18.81	5.575
ViewCrafter [32]	0.291	19.38	0.513	29.85	3.337	2.043	0.457	16.33	0.538	37.92	3.182	2.459
Ours	0.219	23.17	0.762	22.87	1.897	0.890	0.256	20.13	0.741	33.12	2.186	1.198

TABLE I: Quantitative comparison on Tanks-and-Temples [26], RealEstate10K [24], and CO3D [25]. In addition to image-level metrics, R_{dist} and T_{dist} report COLMAP-based reconstruction diagnostics for generated sequences.

experiment should be interpreted as a downstream utility test: the method is not a new 3DGS optimizer, but a sampling strategy that produces observations better aligned with the

fixed reconstruction pipeline.

Table III reports additional results on DTU and Spring. DTU evaluates cross-domain novel view synthesis under a different

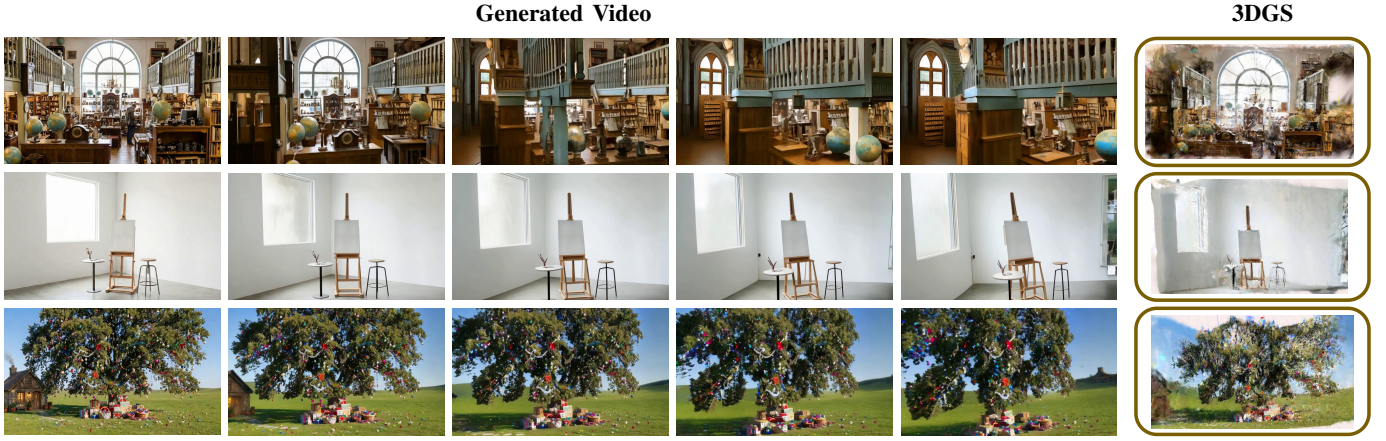


Fig. 4: **Generated views and downstream 3DGS reconstruction.** The generated view sequence is used as observations for downstream 3DGS optimization. GeoConDiffusion produces views that lead to more stable reconstructed geometry under the same reconstruction protocol, especially in regions where invalid cross-view information propagation can produce drift or duplicated structures.

Tanks-and-Temples			
Method	LPIPS ↓	PSNR ↑	SSIM ↑
DNGaussian [39]	0.331	15.47	0.541
FSGS [40]	0.364	17.53	0.558
InstantSplat [34]	0.275	18.61	0.614
ViewCrafter [32]	0.263	19.87	0.699
Ours	0.190	23.45	0.780

ACID			
Method	LPIPS ↓	PSNR ↑	SSIM ↑
pixelNeRF [35]	0.275	18.61	0.614
pixelSplat [3]	0.331	15.47	0.541
MVSplat [4]	0.364	17.53	0.558
ViewCrafter [32]	0.274	18.87	0.701
Ours	0.193	21.45	0.784

TABLE II: Downstream 3D reconstruction results on Tanks-and-Temples [26] and ACID [29]. Generated views are used as observations for the same downstream 3DGS reconstruction protocol.

DTU (cross-domain novel view synthesis)					
Method	PSNR ↑	SSIM ↑	LPIPS ↓	DISTS ↓	FID ↓
pixelSplat [3]	12.89	0.382	0.560	-	-
MVSplat [4]	13.94	0.473	0.385	-	-
TranSplat [5]	14.93	0.531	0.326	-	-
DepthSplat [23]	15.98	0.587	0.398	0.277	131.56
Ours	16.94	0.631	0.263	0.139	85.41

Spring (stereo-view generation)					
Method	PSNR ↑	SSIM ↑	LPIPS ↓	DISTS ↓	FID ↓
ViewCrafter [32]	18.39	0.513	0.272	0.193	144.52
Ours	21.93	0.689	0.194	0.178	138.76

TABLE III: Additional quantitative results on DTU [28] and Spring [30]. DTU evaluates cross-domain novel view synthesis, and Spring evaluates stereo-view generation.

capture setting, while Spring evaluates stereo-view generation where cross-view correspondence is critical. On DTU, GeoConDiffusion improves PSNR and SSIM while reducing LPIPS, DISTS, and FID among the evaluated methods. On Spring, it improves PSNR from 18.39 to 21.93 and SSIM from 0.513 to 0.689 compared with ViewCrafter. These results indicate that the same sampling-time validity mechanism can be beneficial beyond the main datasets, although the protocol differences across methods should be considered when interpreting cross-domain comparisons.

C. Qualitative Results

We further provide qualitative comparisons to examine how the proposed sampling constraints affect local artifacts, long-range stability, and downstream reconstruction behavior. The qualitative analysis follows the same logic as the method design. We first visualize scaffold-related evidence and reconstruction feedback in Fig. 3. We then compare indoor and outdoor novel-view synthesis results in Figs. 5 and 6.

Additional cross-domain and stereo-view examples are shown in Figs. 7 and 8. These visual results are used to analyze typical failure modes rather than to replace the quantitative evaluation.

Figure 5 compares indoor novel-view synthesis results on RealEstate10K. Indoor scenes are sensitive to small geometric deviations because walls, floors, doors, and furniture edges provide strong structural cues. Reconstruction-based methods may produce blurred or incomplete disoccluded regions, while generative methods may synthesize locally plausible but view-inconsistent textures. By using scaffold-derived validity cues during sampling, GeoConDiffusion reduces invalid cross-view feature propagation and better maintains the spatial arrangement of dominant scene structures.

Figure 6 shows outdoor examples on DL3DV-10K. Outdoor scenes often include large background structures and high-frequency regions such as vegetation, tiles, and building facades. These regions are prone to texture swimming when video-diffusion attention propagates features without geometric validity checks. GeoConDiffusion preserves sharper details while maintaining more stable spatial relationships across generated views.

Figure 7 provides additional cross-domain examples on



Fig. 5: **Visual comparisons on RealEstate10K [24].** Indoor scenes contain strong perspective cues, repeated structures, and frequent occlusion boundaries. GeoConDiffusion better preserves global layout and straight structures under viewpoint changes, while reducing blur, ghosting, and local misalignment.

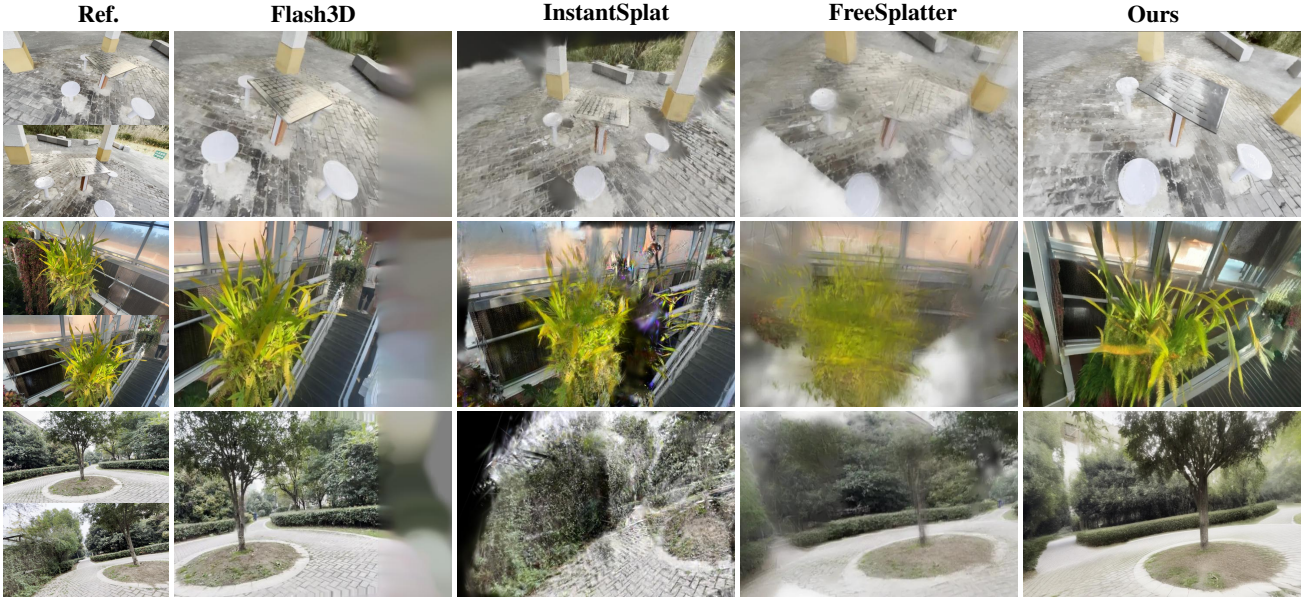


Fig. 6: **Visual comparisons on DL3DV-10K [27].** Outdoor scenes require stable large-scale structures and consistent high-frequency details across camera motion. GeoConDiffusion reduces view-dependent texture drift while preserving sharper local details.

ACID. The results are consistent with the quantitative trends in Table II: GeoConDiffusion better preserves object permanence and local structure under camera motion. The improvement is most visible around thin structures and high-frequency regions, where coarse geometry can provide useful validity evidence but should not be treated as a hard rendering target.

Figure 8 shows stereo-view generation results on Spring. Stereo-view generation is sensitive to local correspondence errors because inconsistent hallucination can directly disrupt binocular matching. The proposed sampling constraints reduce texture drift and view-dependent artifacts, suggesting that geometry-derived validity cues can also help in settings

where correspondence stability is important. This experiment is treated as an additional analysis rather than the main evidence for the method.

D. Ablation Study

We conduct ablation studies to isolate the effect of each sampling constraint. All variants use the same input views, target camera poses, frozen video-diffusion backbone, coarse scaffold, and evaluation protocol. Only the specified sampling component is removed or modified. This design separates the contribution of the proposed inference-time constraints from changes in training data, model architecture, or camera control.

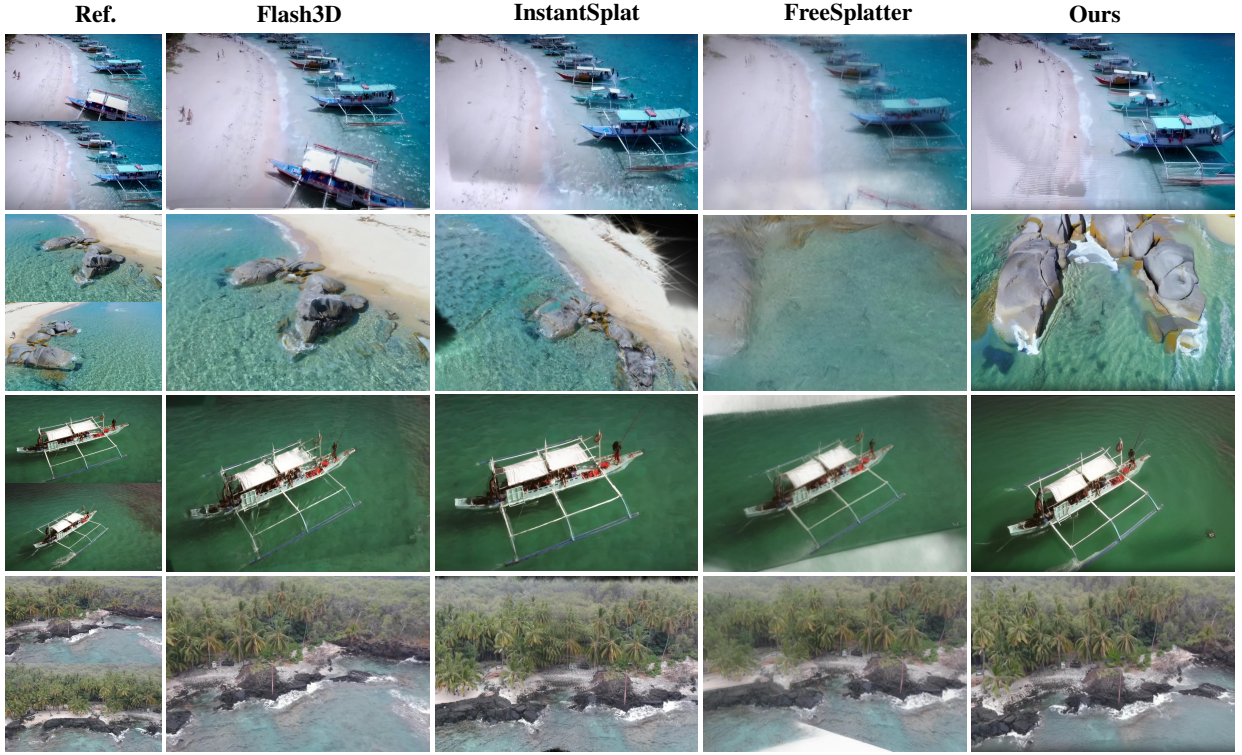


Fig. 7: **Visual comparisons on ACID [29].** GeoConDiffusion maintains object permanence and structural details during viewpoint changes, especially in regions where invalid cross-view inheritance can cause blur or ghosting.

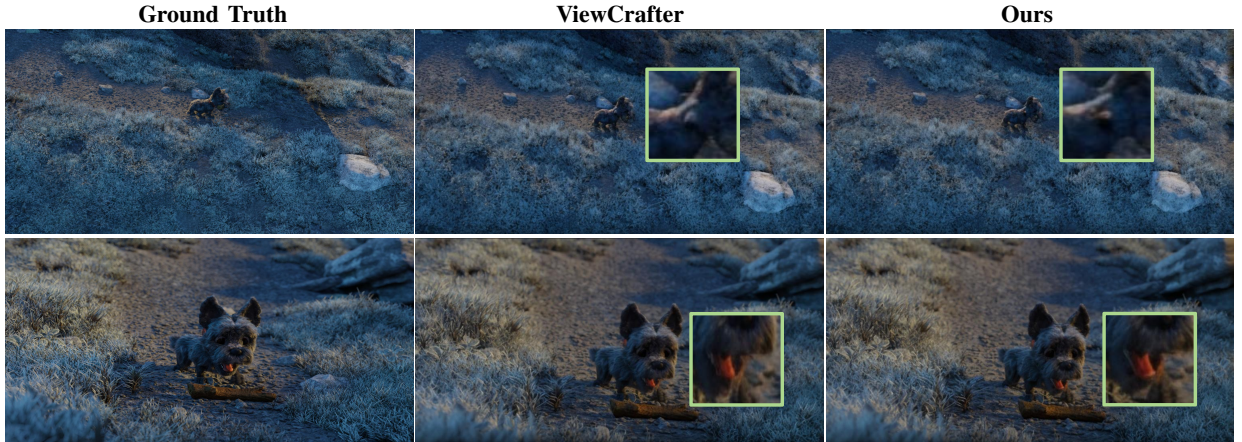


Fig. 8: **Stereo-view generation results on Spring [30].** GeoConDiffusion reduces view-dependent artifacts and texture drift compared with the baseline output, resulting in more stable stereo correspondence under the evaluated setting.

Figure 9 illustrates long-range behavior across generated trajectories. Baseline methods can preserve local plausibility while gradually drifting in object position, boundary sharpness, or background structure. GeoConDiffusion produces more stable trajectories because FMI anchors reliable low-frequency layout at the start of denoising and CGA reduces feature propagation from geometrically unsupported views during denoising.

Table IV reports the quantitative ablation results. Removing FMI causes the largest degradation, indicating that unconstrained Gaussian initialization weakens early layout stabilization and makes the denoising trajectory more prone to global drift. Removing only the spectral mixing component also degrades performance, which suggests that directly relying

Variant	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours (Full)	22.31	0.786	0.132
w/o FMI	20.45	0.712	0.185
w/o CGA	21.08	0.735	0.164
w/o spectral mixing in FMI	21.56	0.751	0.153

TABLE IV: **Ablation study of inference-time sampling constraints.** All variants are evaluated under the same input views, target camera poses, frozen backbone, scaffold, and evaluation protocol. FMI denotes Frequency-Mixed Initialization, and CGA denotes Confidence-Gated Cross-View Attention.

on warped scaffold evidence is insufficient. The initialization should anchor reliable low-frequency structure while leaving high-frequency details to stochastic generation. Removing

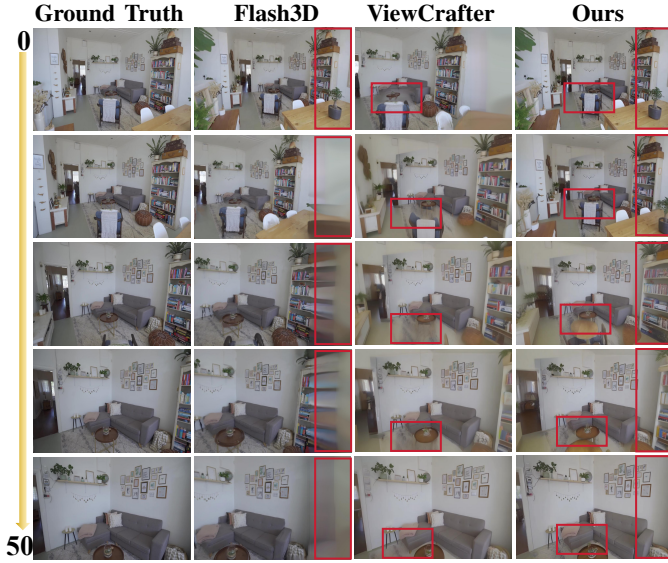


Fig. 9: **Long-range consistency analysis.** The highlighted regions show typical failures under extended viewpoint changes, including object drift, blurred boundaries, and unstable background structures. GeoConDiffusion reduces these artifacts by constraining both the initial latent state and cross-view information flow during sampling.

CGA also reduces performance, confirming that cross-frame attention can still propagate invalid information even when the initial latent is geometrically anchored.

The ablation supports the intended division of roles between the two components. FMI acts before denoising and controls the low-frequency starting point of the sampling trajectory. CGA acts during denoising and controls whether cross-view information is locally valid for target regions. Their combination is therefore not a redundant use of the same scaffold evidence, but two interventions at different stages of the sampling process.

V. CONCLUSION

We presented GeoConDiffusion, an inference-time geometry-constrained sampling strategy for video-diffusion-based novel view synthesis. Instead of treating coarse geometry as a deterministic rendering target, GeoConDiffusion uses scaffold-derived visibility and depth agreement as uncertain validity evidence for sampling-time information inheritance. Frequency-Mixed Initialization anchors reliable low-frequency layout cues at the start of denoising, and Confidence-Gated Cross-View Attention reduces geometrically invalid cross-view feature propagation during denoising. Both components operate on a frozen video-diffusion NVS backbone without denoiser retraining.

Experiments under matched camera protocols show improved image fidelity, perceptual quality, reconstruction-based consistency diagnostics, and downstream 3DGS reconstruction stability. These results indicate that coarse geometry is more defensibly used to regulate where cross-view information can be inherited than to impose a hard rendering target. The method remains limited by the quality of the coarse scaffold

and assumes static scenes with meaningful visibility and depth-ordering cues. Future work may incorporate uncertainty-aware scaffold refinement and more precise correspondence-aware attention control for challenging disocclusions and long camera trajectories.

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